

AI Data Management App - Detailed Fast Track Roadmap

Objective: Become capable of building AI-powered data management applications step by step.

Phase 1 – Database Foundations (Week 1-2)

- 1 Understand what a database is and why it is used
- 2 Install SQLite and DB Browser for SQLite
- 3 Learn SQL commands: CREATE TABLE, INSERT, SELECT, UPDATE, DELETE
- 4 Practice filtering with WHERE and sorting with ORDER BY
- 5 Create tables for users and contacts
- 6 Mini Project: Python Contact Manager using SQLite

Phase 2 – Backend API with FastAPI (Week 3-4)

- 1 Install FastAPI and Uvicorn
- 2 Create first API endpoint
- 3 Understand GET and POST requests
- 4 Connect FastAPI to SQLite database
- 5 Create CRUD endpoints
- 6 Test APIs using browser or Postman

Phase 3 – Frontend Basics (Week 5-6)

- 1 Learn basic HTML structure
- 2 Create forms and tables
- 3 Style pages using CSS
- 4 Use JavaScript Fetch API
- 5 Connect frontend to backend
- 6 Mini Project: Admin Dashboard

Phase 4 – AI Fundamentals (Week 7-8)

- 1 Understand what AI and LLMs are
- 2 Create OpenAI account and API key
- 3 Send prompts to OpenAI API
- 4 Generate summaries from data
- 5 Create AI search feature

Phase 5 – Full AI Data App (Week 9-10)

- 1 User authentication

- 2 Upload files
- 3 Store data
- 4 AI chat with stored data
- 5 Export reports

Phase 6 – Deployment (Week 11-12)

- 1 Install Docker
- 2 Create Dockerfile
- 3 Deploy on VPS
- 4 Configure Nginx
- 5 Add HTTPS certificate

Recommended Daily Study Time: 2-3 hours